

Multi-contrast MRI Reconstruction from Undersampled Data

I. Background

In clinical MRI, reducing scan time is crucial for patient comfort and reducing motion artifacts. However, accelerating MRI scans by undersampling in k-space leads to aliasing artifacts in reconstructed images. This project aims to use the BraTS (Brain Tumor Segmentation) dataset to develop a deep learning-based reconstruction algorithm. The goal is to recover high-quality, aliasing-free images from undersampled data, leveraging both single-modality priors and multi-modal information (e.g., using a fully sampled T1 image to assist T2 reconstruction).

II. Dataset Source:

Public BraTS dataset in NIfTI format (.nii.gz) can be accessed on the 云桌面 (Home/Downloads/dataset) along with demo codes (Home/Desktop) for reading the data.

Modalities: Each patient record includes four co-registered 3D modalities: T1, T1ce, T2, and FLAIR. Only T1 and T2 modalities are used in this project.

Format: For training, the 3D volumes should be processed into 2D slices.

III. Project Assignments

Task 1: Synthesis of Undersampled MRI and Visualization (Level: Easy)

Objective: Master the simulation of MRI undersampling in the Fourier domain and understand the resulting artifacts.

Sampling Pattern Generation: Generate a 2D random variable-density undersampling mask with an acceleration factor of 5.

K-space Simulation: Convert fully sampled **T2 images** to k-space using Fast Fourier Transform. Apply the mask to obtain undersampled k-space data.

Aliasing Implementation: Reconstruct the aliased image using Inverse FFT.

Visualization: Display the sampling mask, the fully sampled image, and the corresponding aliased image side-by-side. Compare the visual differences and aliasing patterns.

Deliverables: Example visualization results of the undersampling process and the generated aliased/fully-sampled image pairs.

Task 2: Baseline Reconstruction Model and Training (Level: Moderate)

Objective: Implement a basic deep learning reconstruction network to remove aliasing artifacts.

Architecture Implementation: Use PyTorch or TensorFlow to build a standalone denoising/reconstruction network (e.g., U-Net, ResNet or DenseNet).

Loss Function Design: Implement L2 Loss (Mean Squared Error) to minimize the pixel-wise difference between the reconstructed image and the fully sampled ground truth.

Model Training: Split the dataset into training, validation, and testing sets with self-defined ratio. Set a reasonable Batch Size and implement a learning rate decay strategy (e.g., ReduceLROnPlateau).

Quantitative Evaluation: Calculate PSNR and SSIM for each testing slice before and after reconstruction.

Deliverables: Model definition code, training logs (Loss curves), and visual comparisons of reconstruction results.

Task 3: Multi-modal Fusion and Advanced Reconstruction (Level: Challenging)

Objective: Introduce advanced strategies, such as multi-modal informed reconstruction and data consistency, to achieve clinical-grade quality.

Architecture Upgrade: Implement an unrolled reconstruction network by cascading the denoising network with a Data Consistency (DC) layer to ensure k-space fidelity.

Multi-modal Informed Reconstruction: Redesign the network to take two modalities as input: an undersampled T2 image and a fully sampled T1 image. Analyze how the structural information from T1 helps recover T2 details.

Loss Optimization: Experiment with L1 Loss or hybrid losses to evaluate whether the reconstructed edge sharpness can be improved.

Error Analysis: Identify the 5 worst-performing reconstruction cases (e.g., significant residual artifacts or lost tumor details) and propose potential improvements.

Deliverables: A complete project code, model weights and a final experimental report.

IV. Submission Guidelines

Deadline (Week 11): Presentation, May 13th; Report, May 17th

The code, report (in PDF format) and presentation slides need to be uploaded to the Blackboard system before the deadlines. Failure to submit your code will result in a 40% reduction of the total assignment mark. Please **claim your use of AI** in this project in the presentation slides.

Only **one** group member is required to submit the required materials for the entire team.

Remember to list the Chinese names of all group members in the report.

V. Grading Policy

Evaluation Item	Weight	Evaluation Criteria
Data Simulation (Task 1)	10%	Correct k-space undersampling and high-quality visualizations.
Model Implementation (Task 2)	25%	Correct network structure, proper training setup, and successful convergence.
Performance Breakthrough (Task 3)	30%	Meaningful multi-modal fusion and unrolled iterations with improved PSNR/SSIM.
Report	15%	Complete charts, rigorous analysis of metrics, and insightful discussion.
Presentation	20%	Clear introduction of the group division of labor, methods, innovations, results, discussion and use of AI in this project.